

Data Cleaning Sheet

Step	Task	Python Function
1	Load Dataset	<code>pd.read_csv()</code>
2	Check Shape	<code>df.shape</code>
3	View Data	<code>df.head()</code>
4	Check Data Types	<code>df.dtypes</code>
5	Dataset Information	<code>df.info()</code>
6	Statistical Summary	<code>df.describe()</code>
7	Find Missing Values	<code>df.isnull().sum()</code>
8	Remove Missing Values	<code>df.dropna()</code>
9	Fill Missing Values	<code>df.fillna()</code>
10	Check Duplicate Rows	<code>df.duplicated().sum()</code>
11	Remove Duplicates	<code>df.drop_duplicates()</code>
12	Rename Columns	<code>df.rename()</code>
13	Change Data Types	<code>astype()</code>
14	Detect Outliers	Box Plot, IQR
15	Handle Outliers	IQR Method
16	Check Unique Values	<code>df.nunique()</code>
17	Encode Categorical Data	LabelEncoder
18	Feature Scaling	StandardScaler
19	Correlation Analysis	<code>df.corr()</code>
20	Save Clean Data	<code>df.to_csv()</code>

Complete Data Cleaning Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("data.csv")

# View first rows
print(df.head())

# Shape
print(df.shape)

# Information
print(df.info())

# Statistical summary
print(df.describe())

# Missing values
print(df.isnull().sum())
```

```
# Fill missing values
df["Age"] = df["Age"].fillna(df["Age"].mean())

# Remove duplicates
df = df.drop_duplicates()

# Rename column
df.rename(columns={"Salary": "Monthly_Salary"}, inplace=True)

# Change datatype
df["Age"] = df["Age"].astype(int)

# Save cleaned data
df.to_csv("cleaned_data.csv", index=False)

print("Data Cleaning Completed")
```

Missing Value Handling

Mean

```
df["Age"].fillna(df["Age"].mean(), inplace=True)
```

Median

```
df["Age"].fillna(df["Age"].median(), inplace=True)
```

Mode

```
df["City"].fillna(df["City"].mode()[0], inplace=True)
```

Remove Missing Rows

```
df.dropna(inplace=True)
```

Duplicate Handling

Find Duplicates

```
df.duplicated().sum()
```

Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

Outlier Detection

Box Plot

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df["Salary"])
plt.show()
```

IQR Method

```
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR

outliers = df[(df["Salary"] < lower) | (df["Salary"] > upper)]

print(outliers)
```

Feature Scaling

Standardization

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[["Salary"]] = scaler.fit_transform(df[["Salary"]])
```

Normalization

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["Salary"]] = scaler.fit_transform(df[["Salary"]])
```

Encoding Categorical Data

Label Encoding

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df["Gender"] = le.fit_transform(df["Gender"])
```

One Hot Encoding

```
df = pd.get_dummies(df, columns=["City"])
```

Real Interview Questions

Q1. Why do we clean data?

To improve data quality and model performance.

Q2. What are missing values?

Values that are not available in the dataset.

Q3. Difference between Mean and Median Imputation?

- Mean: Average value
- Median: Middle value (better for outliers)

Q4. What are outliers?

Extreme values significantly different from other observations.

Q5. Why remove duplicates?

Duplicates can bias analysis and model training.

Q6. What is feature scaling?

Bringing features to a similar range for ML algorithms.

Q7. What is Label Encoding?

Converting categorical values into numerical labels.

Q8. What is One-Hot Encoding?

Creating separate binary columns for each category.

Typical Data Cleaning Workflow

1. Load Data
2. Understand Data
3. Handle Missing Values
4. Remove Duplicates
5. Fix Data Types
6. Detect & Handle Outliers

7. Encode Categorical Variables
8. Scale Features
9. Correlation Analysis
10. Save Clean Dataset